Let's talk about a new topic, "Numerical solution of nonlinear equations".

1 Introduction

Definition 1.1. The function $f(x)$ has a root at $x = r$ if $f(r) = 0$.

We want to find a root of $f(x)$, such that $f(r) = 0$. Let's look some examples. For the first example, let's take a familiar quadratic polynomial as our function, $f(x) = x^2 + 5x + 6$.

Example 1.2. Quadratic polynomials: $f(x) = x^2 + 5x + 6$.

$$f(x) = (x + 2)(x + 3) = 0 \Longrightarrow x_1 = -2, x_2 = -3 \tag{1}$$

- Observation: Roots are not unique.

Example 1.3. $f(x) = x^2 + 4x + 10 = (x + 2)^2 + 6$, no roots.

Example 1.4. The last one is $f(x) = x^2 + \cos x + e^x + \sqrt{x + 1}$. They may not have an explicit expression for the roots.

- Thus the conclusion would be roots can be difficult to find analytically.

Thus in this chapter, we will learn some numerical methods to solve equations such as

- Bisection,
- Fixed point iteration,
- Newton's method,
- Secant method.

For the scope of this lecture, we consider the case when we search for a root of an algebraic equation.

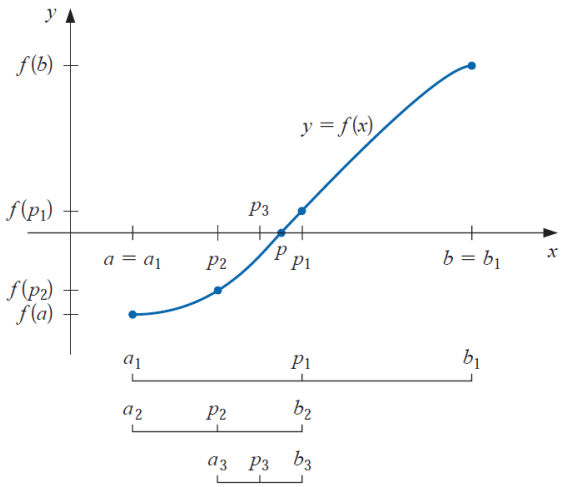
2 The Bisection method

We now look at the bisection method. The basic idea of bisection method is the following.

2.1 Braketing a root

Theorem 2.1. *Let f be a continuous function on $[a, b]$, satisfying $f(a)f(b) < 0$. Then f has a root between a and b, that is, there exists a number r satisfying $a < r < b$ and $f(r) = 0$.*

To see the demonstration of the idea of Bisection on the white board.



Let's summarize it by the following pseudo code.

Bisection Method

Given initial interval $[a, b]$ such that $f(a)f(b) < 0$.

while $(b - a)/2 > TOL$

$c = (a + b)/2$

if $f(c) = 0$, **stop, end**

if $f(a)f(c) < 0$

$b = c$

else

$a = c$

end

end

The final interval $[a, b]$ contains a root.
The approximate root is $(a + b)/2$.

Below is the implementation of this algorithm by MATLAB :

- Click on Editor and click on New to create a new document.
- Type the following piece of code.
- Once you are done, save it by the name `bisect.m`.

```
%Program 1.1 Bisection Method
%Compute approximate solution of f(x)=0
%Input: function handle f; a,b such that f(a)*f(b)<0,
%        and tolerance tol
%Output: Approximate solution xc
function xc=bisect(f,a,b,tol)
if sign(f(a))*sign(f(b)) >= 0
  error('f(a)f(b)< not satisfied!') %ceased execution
end
fa=f(a);
fb=f(b);
while (b-a)/2>tol
  c=(a+b)/2;
  fc=f(c);
  if fc == 0              %c is a solution, done
    break
  end
  if sign(fc)*sign(fa)<0  %a and c make the new interval
    b=c;fb=fc;
  else                    %c and b make the new interval
    a=c;fa=fc;
  end
end
xc = (a+b)/2;             %new midpoint is best estimate
```

The MATLAB you should know to understand this.

- What an m-file is and how the word `function` defines a function stored in a MATLAB file (an m-file)
- The meaning of the decission structures

```
if (conditions)
    ...
end
```

```
if (conditions)
    ...
else
    ...
end
```

```
if (conditions)
    ...
elseif (conditions)
    ...
else
    ...
end
```

  Note that we write `elseif` and not `else if`. The latter means something different.

- How to use a `while` loop.
- How to use `break` to leave a `while` loop.
- The comparison symbol `==`.

Problem 1. Type the following in the command window.

```
>> help bisect
```

What happens?

Problem 2. Please use `bisect.m` to find a root of the function $f(x) = x^3 + x - 1$ in the interval $[0, 1]$ to within 0.5×10^{-10}.

```
>> format long
>> f=@(x) x^3+x-1;
>> xc=bisect( f, 0 ,  1,     0.5*10^(-10)   )
%Please input function handle f and the other 3 parameters.

xc =
```

Now we can use the bisection method to solve an example.

Example 2.2. Find a root of function $f(x) = x^3 + x - 1$ by using the Bisection Method on the interval $[0, 1]$.

Note that all of the roots of the cubic equation can be found algebraically. You can find the general formula for the roots of cubic equation from Wikipedia. For conveniently, we can use the website, http://www.wolframalpha.com/, to get a root by typing $x^3 + x - 1 = 0$. So we now get a exact solution with 15 digits, $x \approx 0.682327803828019$. Can we use the Bisection Method to solve it and get an approximation root? Let's do it.

We compute $f(a_0) = -1, f(b_0) = 1$; and $f(a_0)f(b_0) = (-1)(1) < 0$, so a root exists in the interval $[a_0, b_0]$. The interval mid-point is $c_0 = \frac{1}{2}$. So let's do the first step, $f(c_0) = -\frac{3}{8} < 0$, we know $f(b_0)$ is positive. So we choose the new interval $[a_1, b_1] = [\frac{1}{2}, 1]$. The second step consists of evaluating $f(c_1) = f(3/4) = 11/64 > 0$, leading to the new interval $[a_2, b_2] = [1/2, 3/4]$. Continuing in this way yields the following intervals:

i	a_i	$f(a_i)$	c_i	$f(c_i)$	b_i	$f(b_i)$
0	0.0000	−	0.5000	−	1.0000	+
1	0.5000	−	0.7500	+	1.0000	+
2	0.5000	−	0.6250	−	0.7500	+
3	0.6250	−	0.6875	+	0.7500	+
4	0.6250	−	0.6562	−	0.6875	+
5	0.6562	−	0.6719	−	0.6875	+
6	0.6719	−	0.6797	−	0.6875	+
7	0.6797	−	0.6836	+	0.6875	+
8	0.6797	−	0.6816	−	0.6836	+
9	0.6816	−	0.6826	+	0.6836	+

We conclude from the table that the solution is bracketed between $a_9 \approx 0.6813$ and $c_9 \approx 0.6826$. The midpoint of that interval $c_{10} \approx 0.6821$ is our best guess for the root.

2.2 How accurate and how fast of the Bisection method?

- Consider $[a_0, b_0]$, $c_0 = \frac{a_0+b_0}{2}$, let $r \in (a_0, b_0)$ be a root.

Then we define the error at iteration number 0 which means no bisection.

- Solution errror at iteration number $0 = e_0 = |c_0 - r| \leq \frac{b_0-a_0}{2}$.

  Since we do not know where the root is, we only know the root r lies on the interval from a_0 to b_0. Then we have to think of the worst possible situation. Since r could lie very close to either a_0 or b_0 and c_0 is the midpoint of the interval. So the worst possible value will be half of the length of the interval.

Now we would denote the interval as a_n to b_n for iteration number n.

- Denote the further intervals as $[a_n, b_n]$ for iteration number n.

So for the error at iteration number n, it will equal to $c_n - r$ and then the error estimate for this distance will be again by the same reason, half of the length of the interval, that is, $\frac{b_n-a_n}{2}$.

- $e_n = |c_n - r| \leq \frac{b_n-a_n}{2}$.

We know that the Bisection method after each iteration, the interval length becomes half of the previous one. So a simple induction will show you that this amount $b_n - a_n$ here if you change it into $b_0 - a_0$ for the initial interval. Then you have the division by 2^n. So

- $\frac{b_n-a_n}{2} = \frac{b_0-a_0}{2^{n+1}}$
- Now let's assume we have an error tolerance denoted as ϵ and we require $e_n \leq \epsilon$, then

$$\frac{b_0 - a_0}{2^{n+1}} \leq \epsilon. \tag{2}$$

In order to find the number of iterations needed to achieve the required error tolerance. One would have to solve the inequality below. Let's see how we can do it. So if we take common logarithm on both sides, we have

$$\log_{10}\left(\frac{b_0 - a_0}{2^{n+1}}\right) \leq \log_{10}\epsilon$$
$$\log_{10}(b_0 - a_0) - (n+1)\log_{10} 2 \leq \log_{10}\epsilon$$
$$\log_{10}(b_0 - a_0) - \log_{10}\epsilon - \log_{10} 2 \leq n\log_{10} 2$$
$$n \geq \frac{\log_{10}(b_0 - a_0) - \log_{10} 2\epsilon}{\log_{10} 2} \tag{3}$$

Before we do an example, let's see a definition.

Definition 2.3. A solution is **correct within p decimal places** if the error is less than 0.5×10^{-p}.

Example 2.4. Use the Bisection Method to find a root of $f(x) = \cos x - x$ in the interval $[0, 1]$ to within six correct places.

Step 1. To find how many steps of bisection are required, according to (3), $\epsilon = 0.5 \times 10^{-6}$, then

$$n \geq \frac{\log_{10}(1-0) - \log_{10}(2 \times 0.5 \times 10^{-6})}{\log_{10} 2}$$
$$= \frac{0 - (-6)}{\log_{10} 2} = \frac{6}{\log_{10} 2} \approx \frac{6}{0.301} = 19.9.$$

Therefore, $n = 20$ steps will be needed.

Step 2. Using the Bisection Method to do 20 iterations, we obtain the approximate root to within six correct place is 0.739085.

i	a_i	$f(a_i)$	c_i	$f(c_i)$	b_i	$f(b_i)$
0	0.000000	+	0.500000	+	1.000000	−
1	0.500000	+	0.750000	−	1.000000	−
2	0.500000	+	0.625000	+	0.750000	−
3	0.625000	+	0.687500	+	0.750000	−
4	0.687500	+	0.718750	+	0.750000	−
5	0.718750	+	0.734375	+	0.750000	−
6	0.734375	+	0.742188	−	0.750000	−
7	0.734375	+	0.738281	+	0.742188	−
8	0.738281	+	0.740234	−	0.742188	−
9	0.738281	+	0.739258	−	0.740234	−
10	0.738281	+	0.738770	+	0.739258	−
11	0.738769	+	0.739014	+	0.739258	−
12	0.739013	+	0.739136	−	0.739258	−
13	0.739013	+	0.739075	+	0.739136	−
14	0.739074	+	0.739105	−	0.739136	−
15	0.739074	+	0.739090	−	0.739105	−
16	0.739074	+	0.739082	+	0.739090	−
17	0.739082	+	0.739086	−	0.739090	−
18	0.739082	+	0.739084	+	0.739086	−
19	0.739084	+	0.739085	−	0.739086	−
20	0.739084	+	0.739085	−	0.739085	−

3 Fixed-point iteration

Now we will study another method, Fixed-point iteration, for solving equations.

- Want to solve $f(x) = 0$.
- Rewrite into $x = g(x)$.

Definition 3.1. The real number r is a fixed point of the function g if $g(r) = r$.

Once the equation is written as $g(x) = x$, Fixed-Point Iteration proceeds by starting with an initial guess x_0 and iterating the function g.

Algorithm of Fixed-Point Iteration is summarized below.

Choose x_0

do iteration until stop criteria, $|x_{i+1} - x_i| \leq TOL$

$x_{i+1} = g(x_i)$

end

Back to Example 2.2 Three choices and their corresponding results.

1.
$$x = 1 - x^3 \tag{4}$$

i	x_i
0	0.50000000
1	0.87500000
2	0.33007813
3	0.96403747
4	0.10405419
5	0.99887338
6	0.00337606
7	0.99999996
8	0.00000012
9	1.00000000
10	0.00000000
11	1.00000000
12	0.00000000

2.
$$x = \sqrt[3]{1 - x} \tag{5}$$

3.
$$x = \frac{1 + 2x^3}{1 + 3x^2} \tag{6}$$

Now two equations arise:

Question 1. Why does not the choice of $g(x) = 1 - x^3$ converge?

Question 2. Why dose the choice of $g(x) = \frac{1+2x^3}{1+3x^2}$ converge in a more striking way?

So let us see the answers for these questions in the next section.

i	x_i
0	0.50000000
1	0.79370053
2	0.59088011
3	0.74236393
4	0.63631020
5	0.71380081
6	0.65900615
7	0.69863261
8	0.67044850
9	0.69072912
10	0.67625892
11	0.68664554
12	0.67922234
13	0.68454401
14	0.68073737
15	0.68346460
16	0.68151292
17	0.68291073
18	0.68191019
19	0.68262667
20	0.68211376
21	0.68248102
22	0.68221809
23	0.68240635
24	0.68227157
25	0.68236807

i	x_i
0	0.50000000
1	0.71428571
2	0.68317972
3	0.68232842
4	0.68232780
5	0.68232780
6	0.68232780
7	0.68232780

3.1 Linear convergence of Fixed-Point Iteration

Recall that

$$x_{i+1} = g(x_i), \tag{7}$$

$$r = g(r). \tag{8}$$

We show how it converge. To do a subtraction for the above two equations, we have

$$x_{i+1} - r = g(x_i) - r \tag{9}$$

$$= g(x_i) - g(r) \tag{10}$$

Using the mean value theorem, there exists a number c_i between x_i and r, s.t.

$$x_{i+1} - r = g'(c_i)(x_i - r). \tag{11}$$

We let $e_i = |x_i - r|$, then

$$e_{i+1} = e_i |g'(c_i)|. \tag{12}$$

If $|g'(x)| < S < 1$, we can see

$$e_{i+1} = e_i|g'(c_i)| < e_i S < e_{i-1}S^2 < \cdots < e_0 S^{i+1}. \tag{13}$$

The error decreases to zero as the number of steps increases.

So now we roughly know that if $|g'(r)| < 1$, FPI converges otherwise not. Now the convergence analysis is presented in the following.

Definition 3.2. Let $e_i = |r - x_i|$. If $\lim_{i\to\infty} \frac{e_{i+1}}{e_i} = S < 1$, the method is said to obey linear convergence with rate S.

Theorem 3.3. *Assume that g is continuously differentiable, that $g(r) = r$, and that $S = |g'(r)| < 1$. Then Fixed-Point Iteration converges linearly with rate S to the fixed point r for the initial guesses sufficiently close to r.*

So

The convergence condition of FPI: There exists an interval $I = (r - \epsilon, r + \epsilon)$, for some $\epsilon > 0$, such that $|g'(r)| < 1$ on I and $x_0 \in I$.

Back to Example 2.2

- For $g(x) = 1 - x^3$, the derivative is $g'(x) = -3x^2$, plug in 0.6823 in $g'(x)$

$$S = |g'(r)| = | - 3(0.6823)^2| \approx 1.3966 > 1. \tag{14}$$

  Therefore, this one does not converge.

- For the second choice, $g(x) = \sqrt[3]{1-x}$, the derivative is $g'(x) = \frac{1}{3}(1-x)^{-\frac{2}{3}}(-1)$ and

$$S = |(1 - 0.6823)^{-\frac{2}{3}}/3| \approx 0.716 < 1. \tag{15}$$

  Theorem 3.3 implies convergence, agreeing with our previous calculation.

- For the third choice, $g(x) = \frac{1+2x^3}{1+3x^2}$,

$$\begin{aligned} g'(x) &= \frac{6x^2(1+3x^2) - (1+2x^3)6x}{(1+3x^2)^2} \\ &= \frac{6x(x^3+x-1)}{(1+3x^2)^2} \end{aligned}$$

  and $S = |g'(r)| = 0$.

  This is as small as S we get, leading to the very fast convergence.

Example 3.4. Use Fixed-Point Iteration to find a root of $\cos x = \sin x$.

The simplest way to convert the equation to a fixed-point problem is to add x to each side of the equation. We can rewrite the problem as

$$x + \cos x - sinx = x \tag{16}$$

and define

$$g(x) = x + \cos x - \sin x. \tag{17}$$

The result of applying the Fixed-Point Iteration method to this $g(x)$ is shown in the table.

There are several interesting things to notice in the table. First, the iteration appears to converge to 0.7853982. Since $\cos(\pi/4) = \sqrt{2}/2 = \sin(\pi/4)$, the true solution to the equation $\cos x - \sin x = 0$ is $r = \pi/4 \approx 0.78533982$. The fourth column is the"error column." It shows the absolute value of the difference between the best guess x_i at step i and the actual fixed point r. This difference becomes small near the bottom of the table, indicating convergence toward a fixed point. Notice the pattern in the error column. The errors seem to decrease by a constant factor, each error being somewhat less

i	x_i	$g(x_i)$	$e_i = \lvert x_i - r \rvert$	e_i/e_{i-1}
0	0.0000000	1.0000000	0.7853982	–
1	1.0000000	0.6988313	0.2146018	0.273
2	0.6988313	0.8211025	0.0865669	0.403
3	0.8211025	0.7706197	0.0357043	0.412
4	0.7706197	0.7915189	0.0147785	0.414
5	0.7915189	0.7828629	0.0061207	0.414
6	0.7828629	0.7864483	0.0025353	0.414
7	0.7864483	0.7849632	0.0010501	0.414
8	0.7849632	0.7855783	0.0004350	0.414
9	0.7855783	0.7853235	0.0001801	0.414
10	0.7853235	0.7854291	0.0000747	0.415
11	0.7854291	0.7853854	0.0000309	0.414
12	0.7853854	0.7854035	0.0000128	0.414
13	0.7854035	0.7853960	0.0000053	0.414

than half the previous error. To be more precise, the ratio between successive errors is shown in the final column. In most of the table, we are seeing the ratio e_{k+1}/e_k of successive errors to approach a constant number, about 0.414. In other words, we are seeing the linear convergence relation

$$e_i \approx 0.414 e_{i-1}. \tag{18}$$

This is exactly what is expected, since Theorem 3.3 implies that

$$S = |g'(r)| = |1 - \sin r - \cos r| = \left| 1 - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right| \approx 0.414. \tag{19}$$

Example 3.5. Find the fixed points of $g(x) = 2.8x - x^2$.

The function $g(x) = 2.8x - x^2$ has two fixed points 0 and 1.8, which can be determined by solving $g(x) = x$ by hand, or alternatively, by noting where the graphs of $y = g(x)$ and $y = x$ intersect. Figure shows a cobweb diagram for FPI with initial guess $x = 0.1$. For this example, the iterates

$$\begin{aligned} x_0 &= 0.1000 \\ x_1 &= 0.2700 \\ x_2 &= 0.6831 \\ x_3 &= 1.4461 \\ x_4 &= 1.9579. \end{aligned}$$

and so on, can be read as the intersections along the diagonal.

Even though the initial point $x_0 = 0.1$ is near the fixed point 0, FPI moves toward the other fixed point $x = 1.8$ and converge there. The difference between the two fixed points is the slope of g at $x = 1.8$, given by $g'(1.8) = -0.8$, is smaller than one in absolute value. On the other hand, the slope of g at the other fixed point $x = 0$, the one that repels points, is $g'(0) = 2.8$, which is larger than one in absolute value.

Example 3.6. Which of the following three Fixed-Point Iterations converge to $\sqrt{5}$? Rank the ones that converge from fastest to slowest.

$$(A)\ x \longrightarrow \frac{4}{5}x + \frac{1}{x}\ (B)\ x \longrightarrow \frac{x}{2} + \frac{5}{2x}\ (C)\ x \longrightarrow \frac{x+5}{x+1}.$$

Solution:

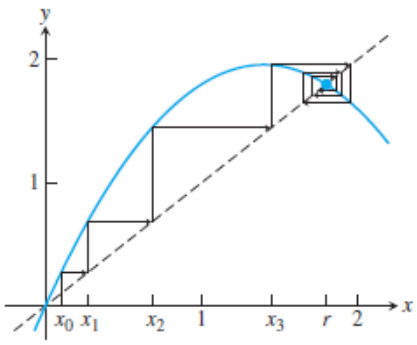


Figure 1.5 Cobweb diagram for Fixed-Point Iteration. Example 1.5 has two fixed points, 0 and 1.8. An iteration with starting guess 0.1 is shown. Only 1.8 will be converged to by FPI.

(A) We consider $g_1 : \mathbb{R} \to \mathbb{R}, x \longrightarrow \frac{4}{5}x + \frac{1}{x}$. First we have $g_1(\sqrt{5}) = \frac{4}{5}\sqrt{5} + \frac{1}{\sqrt{5}} = \sqrt{5}$. So $\sqrt{5}$ is a fixed point of g_1. We compute $g_1'(x)$, get $g_1'(x) = \frac{4}{5} - \frac{1}{x^2}$, therefore $g_1'(\sqrt{5}) = \frac{3}{5}$. We conclude that Fixed Point Iteration on g_1 is locally convergent to the fixed point $r = \sqrt{5}$. The convergence is linear with rate $\frac{3}{5}$.

(B) We consider $g_2 : \mathbb{R} \to \mathbb{R}, x \longrightarrow \frac{x}{2} + \frac{5}{2x}$. First we have $g_2(\sqrt{5}) = \frac{\sqrt{5}}{2} + \frac{5}{2\sqrt{5}} = \sqrt{5}$. So $\sqrt{5}$ is a fixed point of g_2. We compute $g_2'(x)$, get $g_2'(x) = \frac{1}{2} - \frac{5}{2x^2}$, therefore $g_2'(\sqrt{5}) = 0$. We conclude that Fixed Point Iteration on g_2 is locally convergent to the fixed point $r = \sqrt{5}$. The convergence is superlinear.

(C) We consider $g_3 : \mathbb{R} \to \mathbb{R}, x \longrightarrow \frac{x+5}{x+1}$. First we have $g_3(\sqrt{5}) = \frac{\sqrt{5}+5}{\sqrt{5}+1} = \sqrt{5}$. So $\sqrt{5}$ is a fixed point of g_3. We compute $g_3'(x)$, get $g_3'(x) = \frac{-4}{(x+1)^2}$, therefore $|g_3'(\sqrt{5})| \approx 0.3820$. We conclude that Fixed Point Iteration on g_3 is locally convergent to the fixed point $r = \sqrt{5}$. The convergence is linear with rate 0.3820.

The fastest convergence is with $x \longrightarrow \frac{x}{2} + \frac{5}{2x}$, $x \longrightarrow \frac{x+5}{x+1}$ and then comes $x \longrightarrow \frac{4}{5}x + \frac{1}{x}$.

Recall that in Example 3.5, we find that the choice of $x = \frac{1+2x^3}{1+3x^2}$ can converge in a striking way (superlinear rate).

A **question** arise: **Is there a strategy to find the choice of $x = g(x)$, which could converge in a superlinear rate?**

Yes. We will introduce a strategy called **Newton's method**.

4 Newton's method

We know the general Taylor's formula: Let x and x_0 be real numbers, and let f be $k+1$ times continuously differentiable on the interval between x and x_0. Then there exists a number c between x and x_0 such that

$$f(x) = f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\frac{f'''(x_0)}{3!}(x-x_0)^3+\cdots+\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+\frac{f^{(k+1)}(c)}{(k+1)!}(x-x_0)^{k+1}. \tag{20}$$

If let f be 2 times continuously differentiable on the interval between x and x_0. , Then there exists a number c between x and x_0 such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f^{(2)}(c)}{(2)!}(x - x_0)^2. \tag{21}$$

Since $f(x) = 0$, this equation gives

$$0 = f(x_0) + f'(x_0)(x - x_0) + \frac{f^{(2)}(c)}{(2)!}(x - x_0)^2. \tag{22}$$

Newtonąŕs method is derived by assuming that since $|x - x_0|$ is small, the term involving $(x - x_0)^2$ is much smaller, so

$$x \approx x_0 - \frac{f(x_0)}{f'(x_0)}. \tag{23}$$

Solving for x gives an approximation for the root, which we call x_1. Next, the entire process is repeated, beginning with x_1, to produce x_2, and so on, yielding the following iterative formula:

Newton's Method

$$x_0 = \textbf{initial guess}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \text{ for } i = 0, 1, 2, \cdots .$$

Indeed, the Newton's method is a kind of the Fixed-Point iteration method. In fact, we let

$$g(x) = x - \frac{f(x)}{f'(x)}, \tag{24}$$

and we find the root r is a fixed point of the function $g(x)$.

Back to Example 2.2 again

Example 4.1. Find the Newton's Method formula for the equation $x^3 + x - 1 = 0$.

Since $f'(x) = 3x^2 + 1$, we get the choice of

$$g(x) = \frac{1 + 2x^3}{1 + 3x^2}. \tag{25}$$

Then, the formula is given by

$$\begin{aligned} x_{i+1} =& x_i - \frac{x_i^3 + x_i - 1}{3x_i^2 + 1} \\ =& \frac{2x_i^3 + 1}{3x_i^2 + 1}. \end{aligned}$$

Iterating this formula from initial guess $x_0 = -0.7$ yields

$$\begin{aligned} x_1 =& \frac{2x_0^3 + 1}{3x_0^2 + 1} = \frac{2(-0.7)^3 + 1}{3(-0.7)^2 + 1} \approx 0.1271 \\ x_2 =& \frac{2x_1^3 + 1}{3x_1^2 + 1} \approx 0.9577. \end{aligned}$$

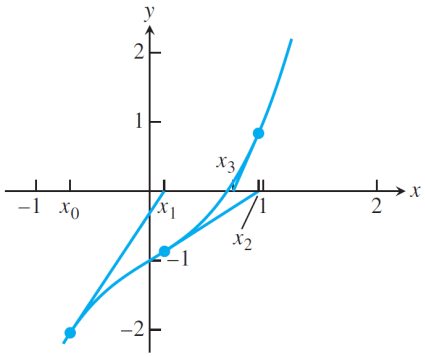


Figure 1.9 Three steps of newton's method. Illustration of Example 1.11. Starting with $x_0 = -0.7$, the Newton's Method iterates are plotted along with the tangent lines. The method appears to be converging to the root.

◀

These steps are shown geometrically in Figure 4. Further steps are given in the following table:

i	x_i	$e_i = \lvert x_i - r \rvert$	e_i / e_{i-1}^2
0	-0.70000000	1.38232780	
1	0.12712551	0.55520230	0.2906
2	0.95767812	0.27535032	0.8933
3	0.73482779	0.05249999	0.6924
4	0.68459177	0.00226397	0.8214
5	0.68233217	0.00000437	0.8527
6	0.68232780	0.00000000	0.8541
7	0.68232780	0.00000000	

After only six steps, the root is known to eight correct digits. There is a bit more we can say about the error and how fast is becomes small. Note in the table that once convergence starts to take hold, the number of correct places in x_i approximately doubles on each iteration. This is characteristic of "quadratically convergent"methods, as we shall see next.

4.1 Quadratic convergence of Newton's Method

The convergence in Example 4.1 is qualitatively faster than the linear convergence we have seen for the Bisection Method and Fixed-Point Iteration. A new definition is needed.

Definition 4.2. Let e_i denote the error after step i of an iterative method. The iteration is **quadratically convergent** if

$$\lim_{i\to\infty} \frac{e_{i+1}}{e_i^2} = M,$$

where

$$M = \frac{f''(r)}{2f'(r)}$$

Theorem 4.3. Let f be twice continuously differentiable and $f(r) = 0$. If $f'(r) \neq 0$, then Newton's Method is **locally** and quadratically convergent to r. The error e_i at step i satisfies

$$\lim_{i\to\infty} \frac{e_{i+1}}{e_i^2} = M, \tag{26}$$

where

$$M = \frac{f''(r)}{2f'(r)}. \tag{27}$$

Proof.

To prove local convergence, note that Newton's Method is a particular form of Fixed-Point Iteration, where

$$g(x) = x - \frac{f(x)}{f'(x)},$$

with derivative

$$g'(x) = 1 - \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2} = \frac{f(x)f''(x)}{f'(x)^2}.$$

Since $g'(r) = 0$, Newton's Method is locally convergent according to Theorem 3.3.

To prove quadratic convergence, we derive Newton's Method a second way, this time keeping a close eye on the error at each step. By error, we mean the difference between the correct root and the current best guess.

Taylor's formula tells us the difference between the values of a function at a given point and another nearby point. For the two points, we will use the root r and the current guess x_i after i steps, and we will stop and take a remainder after two terms:

$$f(r) = f(x_i) + (r - x_i)f'(x_i) + \frac{(r - x_i)^2}{2} f''(c_i).$$

Here, c_i is between x_i and r. Because r is the root, we have

$$\begin{aligned} 0 =& f(x_i) + (r - x_i)f'(x_i) + \frac{(r - x_i)^2}{2} f''(c_i) \\ -\frac{f(x_i)}{f'(x_i)} =& r - x_i + \frac{(r - x_i)^2}{2} \frac{f''(c_i)}{f'(x_i)}, \end{aligned}$$

assuming that $f'(x_i) \neq 0$. With some rearranging, we can compare the next Newton iterate with the root:

$$\begin{aligned} x_i - \frac{f(x_i)}{f'(x_i)} - r =& \frac{(r - x_i)^2}{2} \frac{f''(c_i)}{f'(x_i)} \\ x_{i+1} - r =& e_i^2 \frac{f''(c_i)}{2f'(x_i)} \\ e_{i+1} =& e_i^2 \left| \frac{f''(c_i)}{2f'(x_i)} \right|. \end{aligned}$$

In this equation, we have defined the error at step i to be $e_i = |x_i - r|$. Since c_i lies between r and x_i, it converges to r just as x_i does, and

$$\lim_{i\to\infty} \frac{e_{i+1}}{e_i^2} = \left| \frac{f''(r)}{2f'(r)} \right|,$$

the definition of quadratic convergence. □